

AnyCam 3.0.1-rc1.0 Audit Report

Release audit - 04 October 2026 - appliance cameras, H.265 in the browser, late pictures - tag 3.0.1-rc1.0

Scope: phase 3.0.1 of the build plan, agreed with CrystalHeeler on 2026-10-03, in this order: B2, C10, B20, B23, B24, B26, B12, B8, F10, C11 (also closes B3), C20. CrystalHeeler ordered the phases 3.0.1 to 3.7.0 built one after another without a field test between them, each as its own commit, tag and zip, so that an earlier build can be installed when a later one fails. This build is not field-tested.

1. Cameras in appliances (B2)

Two cameras on test system B never got a card: a litter box camera and a robot vacuum. Both opened no camera port to the scan. Their makers' MAC blocks identify them.

MAC block	Card name	Where the video plays
04:A1:6F:1 (28 bits, iENSO)	Appliance camera (iENSO module)	the appliance maker's app
00:AE:F7 (Dreame)	Dreame robot vacuum	the Dreamehome app

- The scan keeps each live device's MAC address from the ARP sweep (LIVE_HOST_MACS), and the port scan's MAC as a second source.
- Each live device with no camera port is named in the log, with its MAC and maker. Before, it was dropped without a line.
- An appliance camera with no card gets an information card (display 'appliance', status 'info'). One that asks for a login keeps the usual login card, with a note.
- AnyCam does not try the appliances' default passwords or known flaws.

2. H.265 and the browser (C10)

Firefox plays H.265 only on Windows, through Windows itself: it needs Microsoft's HEVC Video Extensions and a graphics chip that decodes H.265. A page cannot install a codec. The page now asks the browser (MediaSource.isTypeSupported with an hvc1 and an hev1 codec string).

Case	What happens
Enhanced View, H.265 stream, browser plays H.265	As before.
... browser cannot play H.265, camera has an H.264 stream	Live view plays the H.264 stream, with a note.
... no H.264 stream	The classic view, with a message: HEVC Video Extensions for Firefox on Windows, else Chrome or Edge.
Live card	The page adds ?h265=0; the server skips H.265 streams (_go2rtc_card_source(h265=False)), or answers why none is left.
A live-view error	Readable text; go2rtc's text stays in the console.

3. Late pictures and insects (B20)

Part	Cause	Change
Cards minutes late	The Pi decodes 3840x2160 H.265 in software slower than the DVR sends it, and works through a backlog.	A 4K H.265 card decodes keyframes only (-skip_frame nokey): about one picture a second, always current.
Old picture on a card	The snapshot endpoint served the last picture at any age.	A picture older than 10 s is not served (503); the card says 'Loading feed'.
Log flood	A failing stream was retried every 10 s with a warning each time.	10, 20, 40, 80, 160, then 300 s; one warning, then debug lines, and one line at recovery.
Insect echo	On the snapshot path a picture compared with an insect picture showed a change.	A picture must also differ from the picture before the reference.

Limit: a one-picture insect on the snapshot path still records, as in 3.0.0. The live stream path, which most cameras use, already needs two changed pictures.

4. The other items

#	Change
B23	Manual add: the protocol name is restored to 'WebRTC' after it is made upper case. WS-RTSP cards get display 'wsrtsp'.
B24	The stream-table check paces by the camera's brand entry (_brand_throttle_seconds), not by a key no STREAM_DB entry has. The pacing label holds no password.
B26	The log link asks /api/self for the add-on's ID, from the Supervisor's /addons/self/info, cached; 'local_camera_discovery' when there is no Supervisor.
B12	VAAPI is used only when the render device exists and ffmpeg opens it (a 0.1 s test encode), tested once per start.
B8	Progress is measured in work done per stage (hosts, ports, ONVIF devices), weighted by the last scan's stage times. The page gets elapsed time and time left.
F10	On the first start AnyCam sets ingress_panel and auto_update through /addons/self/options, once. A failed call sets no marker.
C11, B3	Five settings removed with their code: Low FPS, Skip Non-Reference Frames (invalid 'nonref' value, B3), Limit Threads, Stagger Poll, Fast Stream Start (and the hardware preheater it ran). The Supervisor ignores saved values for removed options.
C20	The Classic button in Enhanced View and its code removed. The classic view still starts when live view cannot play.

Tests run

Suite	Result	Scope
Release gate	PASS	Eight gates; 14 files; build inputs checked online.
Python integration	PASS	471 of 471; 23 new in section AA, and updates in X4, Y2, Y3, Y9, Z4, Z9 and G4.
JavaScript behaviour	PASS	118 of 118, with a C10 section and the C20 check.

Suite	Result	Scope
Undefined names	PASS	14 files, 0.
Not run	NOTED	No field test (CrystalHeeler's order for this run). The image was not built. ffmpeg, the Supervisor, the scan tools and the browser are stand-ins in the tests; -skip_frame nokey and the VAAPI test encode did not run on a Pi.

Best-practices compliance

Measured against *docs/AnyCam_Coding_Best_Practices.md*, Parts 1 to 6. Rows this change touches; the rest carry from the 3.0.0-rc1.5 audit.

Standard	Check applied	Result	Note
1.1 Truncated function	AST, contracts, undefined names	PASS	14 files.
1.3 Runtime errors	Names	PASS	B23 and B24 fixed; removed names gone from every NEEDS list.
1.4 Async	Subprocess and sleep	PASS	The VAAPI test and the Supervisor calls are async; no blocking I/O.
1.10 Exceptions	Failure paths	PASS	Supervisor errors return None; F10 retries at the next start.
1.12 Logging	Volume	PASS	One warning per failing stream instead of one per retry.
1.18 Testing	New tests	PASS	23 server and 7 page checks.
2.x JavaScript	Async, feature detection	PASS	isTypeSupported in try/catch; result cached.
6.1 Security	Credentials	PASS	Pacing label strips the password; no default passwords tried on appliances.
Rule 9 (privacy)	New text	PASS	Tests use made-up addresses and MACs; MAC blocks are makers' public prefixes.

Known issues carried forward

- **B25:** a camera's streams are read only at password entry (3.1.0).
- **B6:** the Microseven waits until CrystalHeeler unlocks it.
- **B20:** a one-picture insect on the snapshot path still records.

Acceptance (field test)

- Test system B: the litter box camera and the robot vacuum have information cards; the log names each skipped device.

- Firefox on Windows without HEVC Video Extensions: the Hikvision PTZ plays its H.264 stream, or shows the message.
- Test system B: 4K channel cards stay current (under 10 s behind).
- The scan bar moves steadily and shows time left.
- The log link opens the add-on's log page.
- After the update, Show in Sidebar and Auto update are on.

Files changed

File	Change
anycam_scan.py	B2 (MACs, dropped-host log, information cards), B8.
anycam_brand.py	B2: appliance MAC blocks.
anycam_snap.py	B20 keyframes only and the 10 s limit; C11 removals.
anycam_motion.py	B20 back-off and second look.
anycam_go2rtc.py	C10: card source without H.265.
anycam_credentials.py	B23, B24.
anycam_focus.py, anycam_page.py	C11, C20 removals.
camera_discovery.py	B26, F10, B12; C11 options; version 3.0.1-rc1.0.
page_script.py	C10, B26, C20, B2 card.
config.yaml, run.sh, translations/en.yaml	C11 options removed; version.
tests/	Section AA; updated checks; a MediaSource fake.
CHANGELOG.md, docs	3.0.1-rc1.0; build plan.